Novel Adaptive Region Spectral–Spatial Features for Land Cover Classification With High Spatial Resolution Remotely Sensed Imagery

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***Abstract*— Spectral–spatial features are important for ground target identification and classification with high spatial resolution remotely sensed (HSRRS) Imagery. In this article, two novel features, named the Gaussian-weighting spectral (GWS) feature and the area shape index (ASI) feature, are proposed to complement the deficiency of the basic image feature for land cover classification with HSRRS imagery. The proposed GWS feature is an adaptive region-based feature that aims to improve the spectral homogeneity of a local area surrounding a pixel. Additionally, it is well known that the spectral feature is inadequate for classifying HSRRS imagery. Therefore, one spatial feature called the ASI feature is proposed here to describe the relationship between the area and shape for an adaptive region around each pixel. The proposed GWS and ASI features coupled with the basic red–green–blue (RGB) feature are fed into a supervised classifier to obtain the final classification map. Experiments based on four real HSRRS images demonstrate that the proposed GWS and ASI features are capable of improving classification accuracies compared with some cognate state-of-the-art methods. Moreover, the experiments also reveal that the proposed spectral–spatial features can complement each other for enhancing the classification performance with HSRRS images.**

***Index Terms*— High spatial resolution remotely sensed (HSRRS) image, land cover classification, spatial feature, spectral feature.**

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I. Introduction

REMOTE sensed images have been widely used for observing the earth's surface, such as change detection [1], [2], [3], [4] and land cover classification [5], [6], [7], [8], [9]. Land cover classification aims to assign a semantic label to each pixel in an image scene. Owing to the quick development of remotely sensed platforms (such as QuickBird satellites, aerial planes, and drones), high spatial resolution remotely sensed (HSRRS) images can be easily obtained [10]. An HSRRS image can capture the ground details, such as shape, boundary, and direction for ground targets, which are very helpful for land cover classification [11], [12], [13], [14], [15]. From the perspective of applications, land cover classification with HSRRS images plays an important role in urban planning [16], target recognition [17], [18], and natural disaster assessment [19], [20].

However, the high spatial resolution does not mean high classification accuracies of HSRRS images [21], [22], [23], [24]. Given that the intraclass variance of HSRRS image is increased and the interclass variance is decreased with the increscent of spatial resolution for remotely sensed images, this phenomenon causes the decrease of the separability for different ground targets while achieving classification with HSRRS image [24]. To solve the problem, various methods have been promoted to improve the performance of classification with HSRRS images, including the spatial–spectral feature extraction methods [11], [25], deep learning techniques [26], feature fusion approaches [27], and post classification approaches [28]. This article concentrates on improving the classification accuracies by extracting spatial features to complement the spectra insufficiency of HSRRS images.

The most popular and intuitive approach for exploring spatial features is to describe the shape and size of ground targets by analyzing the spectral similarity in a spatial continuous area. In this approach, the spectral similarity between a pixel and its neighboring pixels is measured, and the shape of the local area around the pixel is described. For example, length-width extraction algorithm [29], pixel shape index (PSI) [30], new pixel shape feature set [31], and morphological building index [32]. The experimental results with real HSRRS images verified the feasibility and advantages of these algorithms for exploring spatial features and completing the spectral

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insufficiency for classification. However, when an exploration occurs around a noise pixel, the description of the shape and size of a target may be inaccurate, and that will decrease the classification accuracy.

To further deal with the negative affect of the noise in spatial exploration, many object-oriented features have been promoted for classification with HSRRS image [25], [33], such as object-correlative index [34], object-based feature selection [35], and object-based neighbor features [36]. Various object-based classification methods indicated that improving the processed unit from pixel to object can reduce the noise in the classification map with HSRRS images [37]. However, an optical parameter setting for obtaining multiscale objects with a multiscale segmentation algorithm requires trial-and-error experiments and over depends on experience for a given image [38].

Moreover, many mathematical models have been adopted for exploring spatial features and enhancing the classification performance with HSRRS images. The most typical studies of this method are mathematical morphology [39], such as morphological transformations [40], morphological attribute profiles [41], extended morphological profiles (EMPs) [42], and multishape EMPs (MEMPs) [43]. In addition, Markov random field [44], [45], wavelet [46], [47], and level-set [48], [49] have been applied successfully in spatial feature exploration for HSRRS image classification. In these approaches, spatial information is assumed to meet the distribution of a mathematical model. However, a strict mathematical model with a fixed size of a contextual domain cannot cover the ground targets with various shapes and sizes. Aiming to improve the classification performance with HSRRS image, filter processing and feature fusion are also intuitive ways to improve the classification performance by enhancing the spectral homogeneity and feature distinguishability of a ground target. For example, rolling guide filters [50], guided filters [51], [52], and Gabor filters [53] have been widely adopted to improve hyperspectral image classification performance. However, one challenge of applying a filter to processing HSRRS images is improving the spectral homogeneity of a local area that belongs to a ground target and preserving the boundary of different targets simultaneously.

To tackle this challenge, many edge-preserving filters (EPFs) have been promoted to improve classification performance with HSRRS images, such as EPFs [54], [55], [56], iterative EPF [57], and spatial–spectral EPF [58]. In addition, feature fusion methods are suggested to enhance the separability of different targets and improve the classification performance with HSRRS images, such as Liu et al. [59] proposed novel two-stage multiple feature fusion (TsF) approaches for HSRRS image classification, and Zhang et al. [60] proposed a multiscale feature fusion approach for improving remote sensing image classification. Moreover, some researchers suggested to couple image fusion with filtering for improving classification performance with remotely sensed images, such as Kang et al. [61] who proposed a feature extraction approach based on image fusion and recursive filtering (IFRF), and Hu et al. [62] that developed an approach using multiscale feature fusion based on nonlinear diffusion filtering for image classification.

In addition, it worth noting that how to fuse the spatial–spectral features is important for improving classification performance with HSRRS images [63]. Because spatial feature and spectral feature play different importance level in recognizing different ground targets. Such as, it is intuitive that spatial feature is more important in recognizing building than in recognizing meadow. However, lots of studies well verified that exploring spatial–spectral features and fusing them together are feasible and effective for improving remote sensing image application performance [64], [65], [66].

The works discussed above indicated that extracting spatial features and complementing them with spectral features can improve classification performance. However, some challenges of spatial-spectral feature extraction for improving classification performance with HSRRS images remain.

1) Exploring spatial features adaptively for various targets with different shapes and sizes is a challenge. Intuitively, different targets in an image scene, such as buildings, roads, and meadows, usually have different shapes and sizes. Therefore, a mathematical model with a fixed template (such as a structural element of morphological operation) or a regular window is insufficient to fit the targets with various shapes and sizes. How to explore and describe the spatial feature of different targets adaptively remains a challenge for land cover classification with HSRRS images.
2) Smoothening noise to improve the spectral homogeneity of image patches while preserving the details of ground targets is a challenge for HSRRS images. HSRRS image can capture the details of ground targets, but the details may amplify the spectral variance of an image patch that belongs to a ground target. Moreover, spectral noise enhances the intraclass variance and reduces the separability of different entities. Therefore, how to improve the intraclass spectral homogeneity in an HSRRS image while preserving the boundaries among different entities is a challenge.
3) Designing a spatial–spectral feature extraction approach that is robust to different HSRRS images is a challenge. Remote sensing images with high resolution can be easily obtained with different sensors and platforms nowadays, such as satellites, aerial planes, and even unmanned aerial vehicle (UAV). Moreover, different images may have dissimilar characteristics, such as high spatial resolution with red–green–blue (RGB) band images and high spatial resolution with hyperspectral bands images. How to design a universal, robust spatial–spectral feature exploration approach to the different HSRRS images is a challenging task.

In this article, we concentrated on how to explore useful spatial-spectral feature from HSRRS image, our motivation lies in enhancing spectral homogeneity and exploring a spatial feature based on an adaptive region for improving the classification performance of HSRRS images. The major contributions of the article can be briefly summarized as follows.

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1) Two novel features named Gaussian-weighting spectral (GWS) feature and area shape index (ASI) feature are proposed to improve the classification performance with HSRRS images. Because the proposed features are extracted based on an adaptive region around each pixel, they have advantages in improving the homogeneity of different entities and preserving the boundary between two neighboring targets.
2) The proposed GWS and ASI features for classification is robust to different HSRRS images. In particular, the feasibility and robustness of the proposed features are verified by improving the classification performance with HSRRS images, including UAV images, aerial photographs, and hyperspectral images.
3) Advanced classification accuracies can be achieved by coupling our proposed GWS and ASI features. Compared with some typical morphological approaches [42], [43], spatial-spectral approaches [30], EPFs [52], [61], [67], and TsF [59] using four real HSRRS images, our proposed feature achieved the best accuracies in terms of overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Ka) evaluation indicators. In addition, the comparisons between the proposed features and deep learning methods, included deep convolutional neural network (3D-CNN) [68], PresNet [69], adaptive spectral-spatial multiscale network (ASSMN) [70], spectral-spatial fully convolutional network (SSFCN) [71], and S3-Net [72], were conducted in Section III. The improvement achieved by our proposed features is also beneficial for balancing user accuracy for different classes.

The remainder of the article is organized as follows. The details of the proposed approach are presented in Section II, and experiments are given in Section III. Finally, conclusions are drawn in Section IV.

II. Methodology

In this section, the proposed approach is presented in detail. Fig. 1 shows the flowchart of the proposed approach for classification. It contains three parts: 1) generating an adaptive region around each pixel; 2) defining the description of the proposed features; and 3) achieving classification with the proposed features and accuracy evaluation. Details of each part are presented below.

A. Generation of an Adaptive Region Around Each Pixel

Many studies have indicated that utilizing contextual information can enhance the application performance of HSRRS images. In this section, an adaptive region generation algorithm is developed to provide an adaptive contextual constraint for exploring spectral–spatial features around each pixel. Details are presented below. First, assume there is a pixel $P_{i,j}$ with neighboring pixels $P_{m,n}$, where $0 \le i \le W$ and $0 \le j \le H$ (W and H denote the width and height of the image, respectively). In Fig. 2, $P_{m,n}$ denotes one of the eight connected pixels around a central pixel in a searching template. Second, the spectral similarity between a central pixel and its neighboring pixel is used to determine whether a pixel belongs to an adaptive region or not. If a neighboring pixel satisfies two predefined conditions: 1) if $\|P_{i,j} - P_{m,n}\| \le T_1$ and 2) the total number of the assigned pixel of an adaptive region is less than T_2, then $P_{m,n}$ is assigned as the adaptive region, and $P_{m,n}$ is taken as the central pixel for next iterative searching. Finally, if one of the conditions is not satisfied, the extension around $P_{i,j}$ is terminated.

B. Description of Proposed Features

The above statement implies that an adaptive region is composed of a ground of pixels that are spectrally similar and spatial continuously, and an adaptive region is usually a part of a target in an image scene. Under this background, two novel features, named the GWS feature and ASI feature, are proposed based on the adaptive region. Here, assume the pixels within an adaptive region around $P_{i,j}$ are $x_{m,n}$. Details of the two features are calculated by the following equations:

$$\mathrm{GWS}_{i,j,b} = \sum \left(\frac{G\left(\|P_{i,j} - x_{m,n}\|\right)}{W_{i,j}} \times x_{m,n,b} \right), \quad i \neq m,\ j \neq n \tag{1}$$

$$\mathrm{ASI}_{i,j} = \frac{\sqrt{\mathrm{Area}\left(R_{i,j}\right)}}{1 + \sqrt{V(m) + V(n)}}, \quad m \in R_{i,j},\ n \in R_{i,j} \tag{2}$$

where $\mathrm{GWS}_{i,j,b}$ denotes the GWS feature for the pixels around the pixel $P_{i,j}$. $G(\cdot)$ is the normal distribution function, $\|P_{i,j} - x_{m,n}\|$ is the Euclidean distance between the central pixel $P_{i,j}$ and its neighboring pixel $x_{m,n}$, $W_{i,j}$ is the sum of all the Euclidean distance between $P_{i,j}$ and all the pixels within the adaptive region, and "b" denotes the selected band. $\mathrm{ASI}_{i,j}$ denotes the ASI feature, and $\mathrm{Area}(R_{i,j})$ is the area of the adaptive region $R_{i,j}$ around $P_{i,j}$. $V(m)$ and $V(n)$ present the variance for the horizontal coordinate and vertical coordinate of the pixels within the adaptive region, respectively. Within an adaptive region, (1) indicates that the near pixels contribute more effect to the central pixel compared with the distance pixels, and this operation can improve the homogeneity of a local area while preserving the boundary between two neighboring targets, as an example shown in Fig. 3. In addition, (2) reveals the relationship between the area and shape of an adaptive region, and its feasibility and performance will be verified in Section III.

C. Classification and Evaluations

Here, a support vector machine (SVM) classifier is adopted for evaluating the feasibility and performance of the proposed features and the refereed methods. In particular, the fivefold cross-validation and the radial basis function kernel are selected to obtain the final classification maps.

In addition, four widely used evaluation indicators named OA, AA, Ka, and F1 score (F1) are adopted for quantitative companions. Details of each indicator are presented in Table I. In addition, TP and TN denote the true positive and true negative pixels, FP and FN are the false positive and false negative pixels. Moreover, the Standard Deviation User's Accuracy (SDUA) is calculated to access the balancing ability of the proposed approach in terms of the user's accuracy [73].

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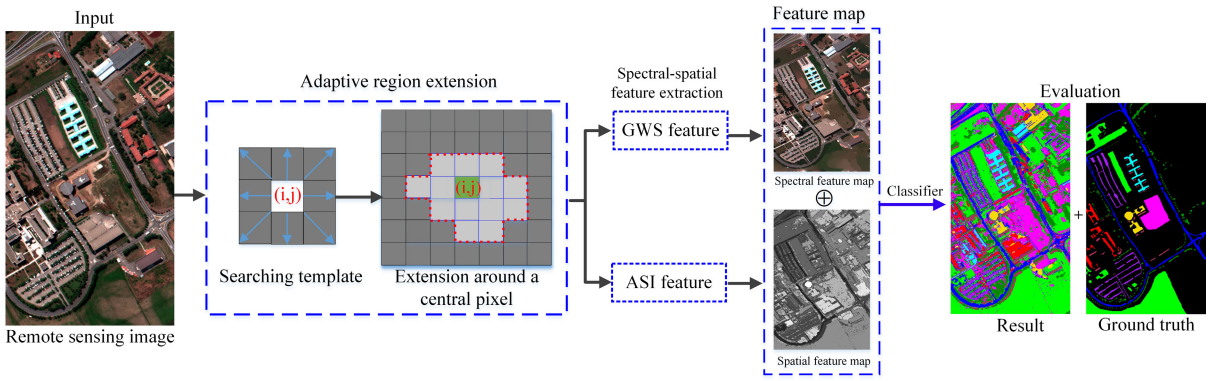


Fig. 1. Flowchart of the proposed features for classification with HSRRS image.

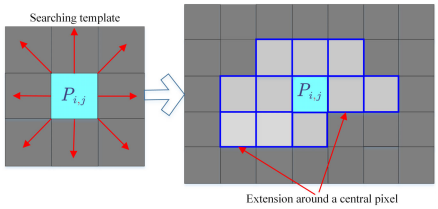


Fig. 2. Schematic illustration of an adaptive region extension around a pixel.

TABLE I
BRIEF SUMMARY OF THE APPLIED EVALUATION INDICATORS

Overall Accuracy (OA)	$OA = \frac{TP+TN}{TP+TN+FP+FN}$
Average Accuracy (AA)	AA = $(\frac{TP}{TP+FP} + \frac{TN}{TN+FN})/T$
Kappa Coefficient (Ka)	$P_0 = \frac{TP+TN}{TP+TN+FP+FN}$, $P_e = \frac{(TP+FP)\times(TP+FN)+(FN+TN)\times(FP+TN)}{(TP+TN+FP+FN)^2}$, $Ka = \frac{P_0-P_e}{1-P_e}$
F1-score (F1-score)	$F1-score = \frac{2TP}{2TP+FN+FP}$
Standard Deviation User's Accuracy (SDUA)	$SDUA = \sqrt{\frac{1}{T}\sum_{i=1}^{i=T}\left(UA_i - \overline{UA}\right)^2}$

Where UA_i is the user's accuracy for ith class, and $\overline{UA}$ is the mean value for all the user's accuracies, and T is the total number of classes refereed in an image scene. A low value of SDUA indicates that the user's accuracy tends to be close to the mean of all the classes, while a high value of SDUA indicates that the user's accuracies are spread out over a wider range.

III. EXPERIMENTS

In this section, the feasibility and superiority of the two proposed features were verified by classification with HSRRS image. Seven typical spatial–spectral feature extraction methods and five state-of-the-art deep learning methods were compared with the proposed approach to demonstrate the performance and superiorities of the proposed features. Details of the experiments are presented below.

A. Data Description

Four real HSRRS images were adopted in the experiments. Table II summarizes that these datasets were obtained with different platforms, including aerial planes with ROSIS sensors and UAVs with Canon 5D-Mark-II cameras. The datatypes of the four datasets contained RGB-false color images with three bands and hyperspectral images, and the spatial resolution included 0.1, 0.32, and 1.3 m/pixel.

In addition, the visual performance and the ground reference map are shown in Figs. 4–7. The experimental datasets covered different scenes, including rural area, suburban area, and urban center area. Moreover, many targets are similar in terms of spectral reflectance, such as grass and trees, road, and buddings. Therefore, classifying the image into a semantic map for each image scene is a challenge.

B. Experiment Setting

To achieve the experiment objectives, the parameter of each approach was optimized by trial-and-error experiments. Details of the experimental setting are given as follows.

1) Training Sample Selection: For all the datasets and each method, 50 pixels with semantic labels were selected randomly from the ground reference map, and each approach used the same training samples for classification to guarantee comparative fairness.
2) Parameters of Each Method.
   a) PSI [30]: $T_1 = 40$, $T_2 = 100$, $D = 20$.
   b) EMPs [42]: SE is a "disk" with an increasing size: "$2 \times 2, 4 \times 4, 6 \times 6$."
   c) MEMPs [43]: SE = "disk," "line," "square," "diamond" with a fixed size of 4×4 pixels.
   d) EPF [54]: $\delta_s = 4$, $\delta_r = 0.2$, $r = 4$, $\in = 0.01$.
   e) IFRF [61]: $\delta_s = 200$, $\delta_r = 30$, integration $= 3$.
   f) RGF [67]: $\delta_s = 4$, $\delta_r = 0.1$, integration $= 5$.
   g) TsF [59]: Intergroup fusion: $K = 7, z, r \in [3, 10]$, $\delta = 0.02$, GFF : $r_1 = 45$, $\delta_1 = 0.3$, $r_2 = 7$, $\delta_2 = 10^{-6}$.
   h) Parameters of the Proposed Approach: T_1 and T_2 are set to 5 and 550 for Data-1 and Data-2, respectively. T_1 and T_2 are set to 7 and 400 for Data-3 and Data-4, respectively.

C. Experimental Results

Based on the preceding parameters settings, the results of the reach approach were compared and presented below. In the

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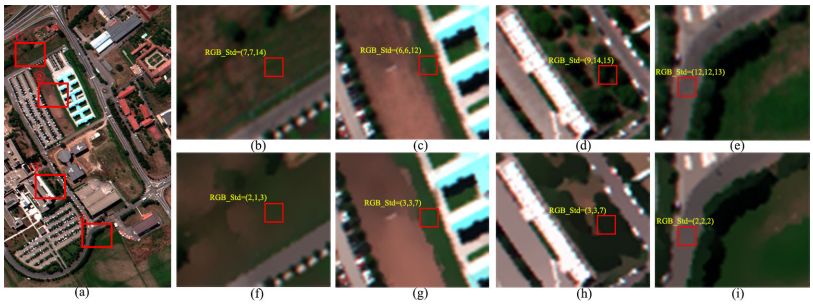


Fig. 3. Comparison between the RGB basic feature and our proposed GWS feature: (a) 1–4 are the observing area with meadows, trees, and roads. (b) and (f), (c) and (g), (d) and (h), and (e) and (i) are the comparative pairs before and after our proposed GWS feature extraction. Notably, RGB_std denotes the standard deviation for the red, green, and blue bands.

TABLE II
DETAIL DESCRIPTION FOR EACH DATASET

Dataset	Location	Platform and Sensors	Size (pixel)	Resolution (m/pixel)
Data-1	China, Jiangxi	UAV and Canon 5D-Mark-II	$1400 \times 100 \times 3$	0.1
Data-2	USA, California	Aerial plane and aerial camera	$560 \times 360 \times 3$	0.32
Data-3	Italy, Pavia University	Aerial planes and ROSIS sensor	$340 \times 610 \times 103$	1.3
Data-4	Italy, Pavia Center	Aerial planes and ROSIS sensor	$715 \times 1096 \times 120$	1.3

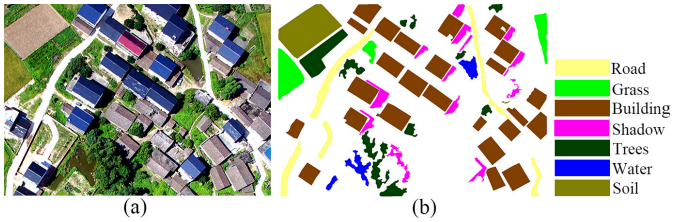


Fig. 4. Dataset-1 (UAV images). (a) Three-band false-color image and (b) ground reference map and class legend.

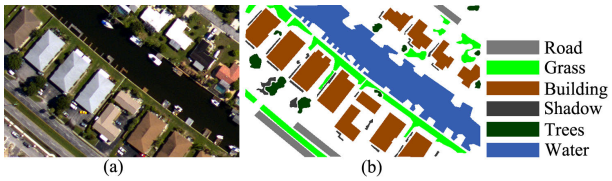


Fig. 5. Dataset-2 (aerial photographs). (a) Three-band false-color image and (b) ground reference map and class legend.

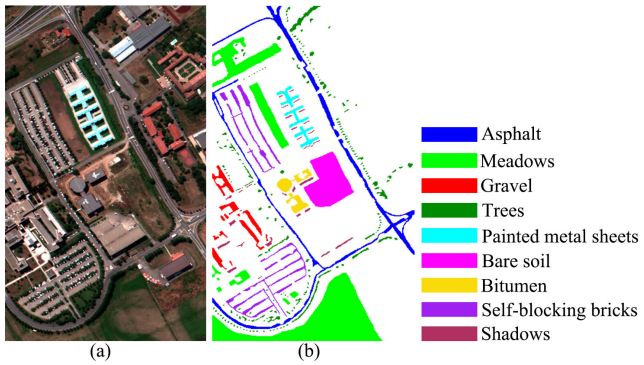


Fig. 6. Dataset-3 (Pavia University, ROSIS hyperspectral image). (a) Three--band false-color image and (b) ground reference map and class legend.

first experiment, the proposed features compared with the spatial features for classifications included PSI [30], EMPs [42], MEMPs [43], EPF [54], IFRF [61], RGF [67], and TsF [59].

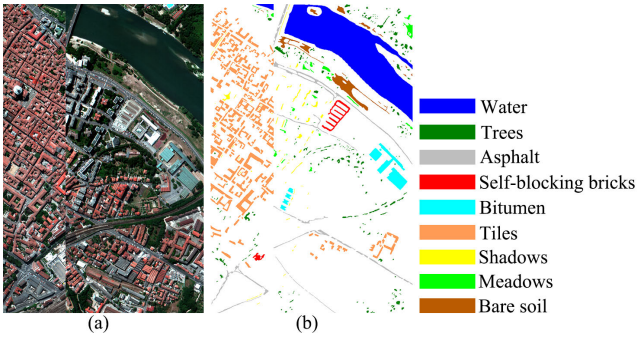


Fig. 7. Dataset-4 (Pavia center, ROSIS hyperspectral image). (a) Three-band false-color image and (b) ground reference map and class legend.

The quantitative comparisons for Data-1–4 are presented in Tables III–VI correspondingly. The comparisons indicated that exploring spatial feature and coupling it with spectral features can improve classification accuracies. For example, EMPs [42] coupled with the original RGB bands achieved better accuracy than using the original RGB-feature only in terms of Data-1. In addition, our proposed feature achieved OA = 98.05%, Ka = 0.973, and F1-score = 97.5% which are the best accuracies compared with other selected methods. Particularly, our proposed approach achieved an AA = 97.85% which is competitive to the best AA = 97.86% which is achieved by EPF [54]. A similar observation can be obtained from the comparisons in Tables IV–VI. The visual performance achieved by each approach is presented in Figs. 8–11. Our proposed approach achieved the classification maps have less salt-and-pepper noise, and the shape and boundary of ground targets are well preserved in the classification maps. The visual comparisons of classification map well support the conclusion of quantitative comparisons.

In addition, the proposed approach was compared with five deep learning methods to further verify the classification performance. As shown in Tables VII–X, the quantitative

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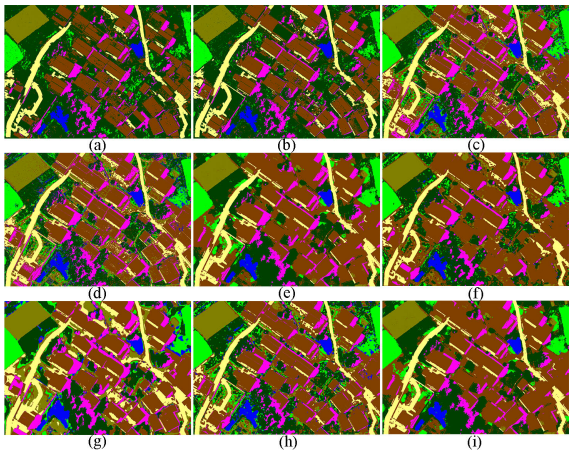


Fig. 8. Acquiring classification maps different traditional methods for Data-1. (a) RGB, (b) PSI [30], (c) EMPs [42], (d) MEMPs [43], (e) EPF [54], (f) TsF [59], (g) IFRF [61], (h) RGF [67], and (i) our proposed approach.

TABLE III

QUANTITATIVE COMPARISON BETWEEN THE SELECTED TRADITIONAL METHODS AND THE PROPOSED APPROACH FOR DATA-1

Feature	OA(%)	AA(%)	Kappa	F1-score(%)	SDUA
RGB	91.09	90.23	0.887	90.85	13.54
PSI [30]	93.87	94.40	0.918	94.48	10.60
EMPs [42]	96.36	95.12	0.951	95.63	4.64
MEMPs [43]	95.32	93.64	0.937	94.27	3.60
EPF [54]	97.88	97.86	0.971	97.43	2.66
TsF [59]	97.09	96.39	0.961	96.69	3.46
IFRF [61]	97.04	95.84	0.960	96.20	5.61
RGF [67]	98.03	97.29	0.973	97.32	**2.00**
Proposed	**98.05**	**97.85**	**0.973**	**97.50**	3.05

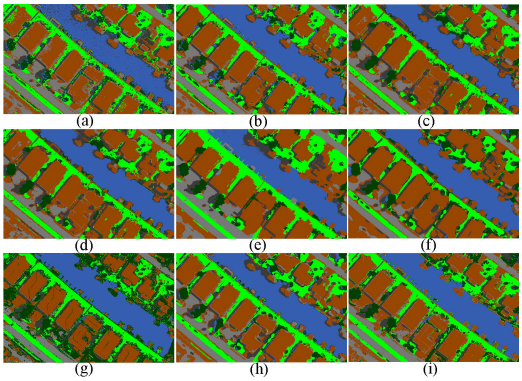


Fig. 9. Acquiring classification maps different traditional methods for Data-2. (a) RGB, (b) PSI [30], (c) EMPs [42], (d) MEMPs [43], (e) EPF [54], (f) TsF [59], (g) IFRF [61], (h) RGF [67], and (i) our proposed approach.

comparisons well demonstrated that our proposed approach has an advantage in improving the classification accuracies while comparing with some state-of-the-art deep learning methods, included 3D-CNN [68], PResNet [69], ASSMN [70], SSFCN [71], and S3-Net [72]. The visual performance observations, presented in Figs. 12–15, supported the quantitative investigated conclusion.

In the second experiment, an experiment was conducted based on different feature compositions for Data-2 and Data-3. Fig. 16 shows two aspects.

1) Compared with the classification using basic RGB feature, the classification accuracies improve when coupled with our proposed GWS or ASI feature. Moreover, the improvement achieved by coupling with the proposed spatial feature ASI is better than that of using proposed GWS. For example, the improvements achieved by the GWS feature are 6.01% and 10.27% for Data-2 and Data-3 in terms of OA, respectively, but the improvements achieved by the ASI feature are 9.08% and 23.4%.
2) Coupling the proposed GWS and ASI features with the basic RGB feature achieved the best classification accuracies compared with individually using GWS or

TABLE IV

QUANTITATIVE COMPARISON BETWEEN THE SELECTED TRADITIONAL METHODS AND THE PROPOSED APPROACH FOR DATA-2 WITH SVM CLASSIFIER

Feature	OA(%)	AA(%)	Kappa	F1-score(%)	SDUA
RGB	88.29	77.03	0.841	80.26	23.83
PSI [30]	93.64	85.97	0.910	87.88	14.33
EMPs [42]	95.26	87.75	0.934	89.62	13.68
MEMPs [43]	94.90	86.63	0.929	89.99	12.98
EPF [54]	93.88	86.45	0.916	88.69	15.17
TsF [59]	95.22	87.40	0.934	90.45	11.53
IFRF [61]	95.91	90.63	0.943	91.95	11.20
RGF [67]	94.36	86.63	0.922	89.00	12.43
Proposed	**98.64**	**96.52**	**0.981**	**96.82**	**2.85**

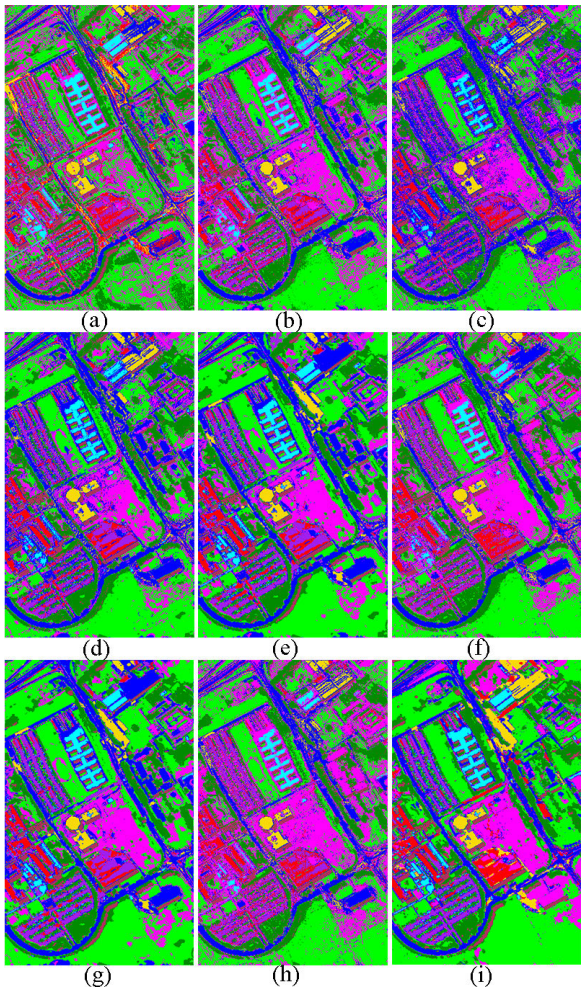


Fig. 10. Acquiring classification maps different traditional methods for Data-3. (a) RGB, (b) PSI [30], (c) EMPs [42], (d) MEMPs [43], (e) EPF [54], (f) TsF [59], (g) IFRF [61], (h) RGF [67], and (i) our proposed approach.

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TABLE V

QUANTITATIVE COMPARISON BETWEEN THE SELECTED TRADITIONAL METHODS AND THE PROPOSED APPROACH FOR DATA-3 WITH SVM CLASSIFIER

Feature	OA(%)	AA(%)	Kappa	F1-score(%)	SDUA
RGB	65.25	67.75	0.570	69.58	22.85
PSI [30]	83.12	78.42	0.782	80.86	12.63
EMPs [42]	84.98	82.06	0.805	83.03	11.60
MEMPs [43]	85.33	81.12	0.809	83.38	10.21
EPF [54]	87.81	84.56	0.841	86.87	9.83
TsF [59]	90.11	85.90	0.871	88.05	11.07
IFRF [61]	88.90	85.89	0.855	87.9	8.33
RGF [67]	88.04	84.77	0.845	86.61	11.43
Proposed	**91.85**	**88.70**	**0.893**	**90.11**	**7.18**

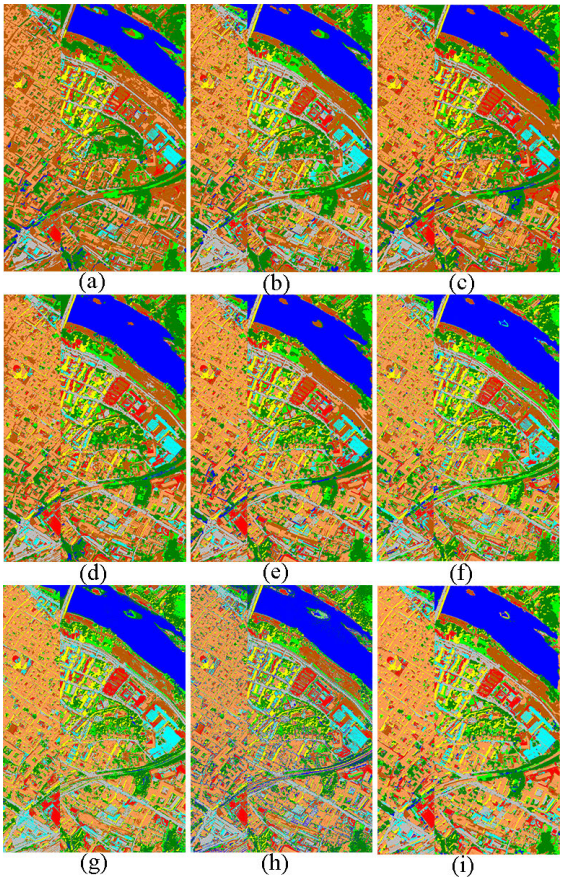


Fig. 11. Acquiring classification maps different traditional methods for Data-4. (a) RGB, (b) PSI [30], (c) EMPs [42], (d) MEMPs [43], (e) EPF [54], (f) TsF [59], (g) IFRF [61], (h) RGF [67], and (i) our proposed approach.

TABLE VI

QUANTITATIVE COMPARISON BETWEEN THE SELECTED TRADITIONAL METHODS AND THE PROPOSED APPROACH FOR DATA-4 WITH SVM CLASSIFIER

Feature	OA(%)	AA(%)	Kappa	F1-score(%)	SDUA
RGB	87.35	77.55	0.826	77.24	23.32
PSI [30]	92.23	86.05	0.893	87.22	17.94
EMPs [42]	93.31	85.85	0.906	87.01	17.82
MEMPs [43]	93.57	86.1	0.911	87.97	16.13
EPF [54]	95.18	86.54	0.932	88.12	15.95
TsF [59]	95.38	88.94	0.935	89.99	13.82
IFRF [61]	94.78	85.74	0.927	88.65	12.14
RGF [67]	92.22	80.37	0.891	83.26	16.25
Proposed	**96.33**	**91.33**	**0.948**	**91.85**	**10.51**

ASI with RGB feature because spatial and spectral features complement each other to enhance the separability among different targets. Furthermore, observing the accuracy based on different feature composition in terms of SDUA can find that our proposed feature not only has advantages in terms of improving classification accuracies but also has a benefit in balancing the user's accuracy for each specific class.

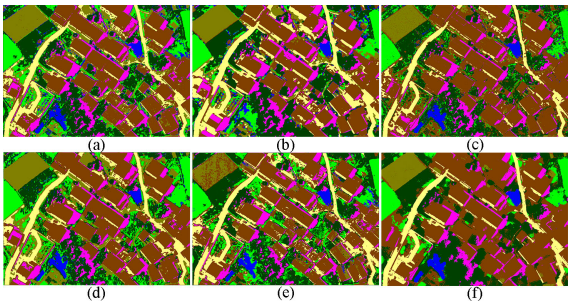


Fig. 12. Visual classification maps acquired with some state-of-the-art deep learning methods and the proposed approach for Data-1. (a) 3D-CNN [68], (b) PResNet [69], (c) ASSMN [70], (d) SSFCN [71], (e) S3-Net [72], and (f) our proposed approach.

TABLE VII

QUANTITATIVE COMPARISON BETWEEN THE SOME STATE-OF-THE-ART DEEP LEARNING METHODS AND THE PROPOSED APPROACH FOR DATA-1

Feature	OA(%)	AA(%)	Kappa	F1-score(%)	SDUA
3D-CNN [68]	96.55	93.96	0.954	93.71	**3.02**
PResNet [69]	94.29	91.93	0.923	89.36	9.00
ASSMN [70]	96.54	96.09	0.953	96.00	3.35
SSFCN [71]	97.15	95.44	0.962	96.18	3.96
S3-Net [72]	91.03	91.06	0.877	90.91	6.25
Proposed	**98.05**	**97.85**	**0.973**	**97.50**	3.05

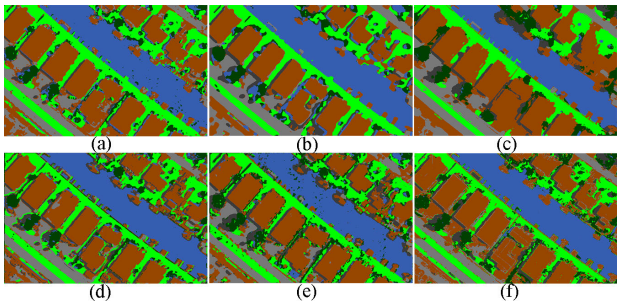


Fig. 13. Visual classification maps acquired with some state-of-the-art deep learning methods and the proposed approach for Data-2. (a) 3D-CNN [68], (b) PResNet [69], (c) ASSMN [70], (d) SSFCN [71], (e) S3-Net [72], and (f) our proposed approach.

TABLE VIII

QUANTITATIVE COMPARISON BETWEEN THE SOME STATE-OF-THE-ART DEEP LEARNING METHODS AND THE PROPOSED APPROACH FOR DATA-2

Feature	OA(%)	AA(%)	Kappa	F1-score(%)	SDUA
3D-CNN [68]	96.84	91.37	0.956	90.54	5.12
PResNet [69]	94.29	91.93	0.923	89.36	9.00
ASSMN [70]	96.37	90.35	0.950	92.14	16.45
SSFCN [71]	96.51	90.37	0.952	93.05	12.87
S3-Net [72]	93.48	85.24	0.910	87.98	18.44
Proposed	**98.64**	**96.52**	**0.981**	**96.82**	**2.85**

D. Discussion

In this section, the relationship between classification accuracies and the parameters of our proposed approach, and the training samples are analyzed to promote its application in practice.

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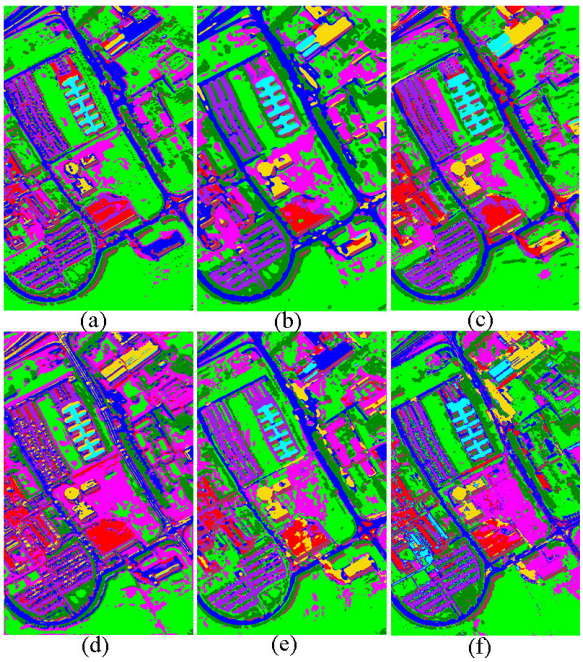


Fig. 14. Visual classification maps acquired with some state-of-the-art deep learning methods and the proposed approach for Data-3. (a) 3D-CNN [68], (b) PResNet [69], (c) ASSMN [70], (d) SSFCN [71], (e) S3-Net [72], and (f) our proposed approach.

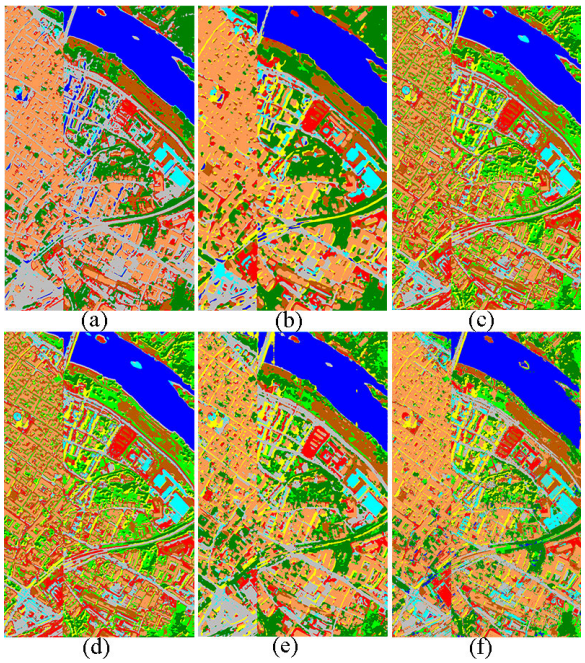


Fig. 15. Visual classification maps acquired with some state-of-the-art deep learning methods and the proposed approach for Data-4. (a) 3D-CNN [68], (b) PResNet [69], (c) ASSMN [70], (d) SSFCN [71], (e) S3-Net [72], and (f) our proposed approach.

TABLE IX
QUANTITATIVE COMPARISON BETWEEN THE SOME STATE-OF-THE-ART DEEP LEARNING METHODS AND THE PROPOSED APPROACH FOR DATA-3

Feature	OA(%)	AA(%)	Kappa	F1-score(%)	SDUA
3D-CNN [68]	82.64	79.11	0.766	72.93	10.42
PResNet [69]	90.66	**90.84**	0.875	87.91	9.42
ASSMN [70]	86.61	87.32	0.823	88.62	12.75
SSFCN [71]	87.01	83.50	0.831	85.03	16.24
S3-Net [72]	88.95	89.87	0.853	89.81	9.46
Proposed	**91.85**	88.70	**0.893**	**90.11**	**7.18**

TABLE X
QUANTITATIVE COMPARISON BETWEEN THE SOME STATE-OF-THE-ART DEEP LEARNING METHODS AND THE PROPOSED APPROACH FOR DATA-4

Feature	OA(%)	AA(%)	Kappa	F1-score(%)	SDUA
3D-CNN [68]	93.09	85.81	0.902	86.31	13.19
PResNet [69]	95.24	86.46	0.932	81.75	12.48
ASSMN [70]	95.30	88.26	0.934	90.28	10.16
SSFCN [71]	94.80	86.03	0.927	88.47	14.35
S3-Net [72]	95.13	86.15	0.932	89.21	10.47
Proposed	**96.33**	**91.33**	**0.948**	**91.85**	**10.51**

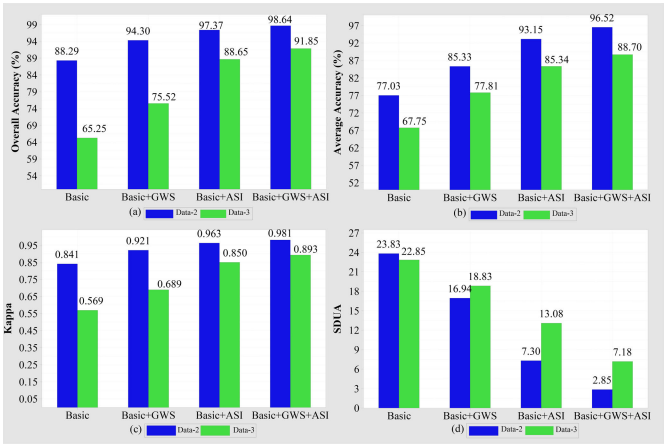


Fig. 16. Comparisons among different feature compositions (Basic indicates the original RGB features, GWF is the Gaussian-weight spectral feature, and ASI feature is area-shape index feature) for different classification accuracy metrics. (a) Overall accuracy, (b) Average accuracy, (c) Kappa coefficient, and (d) SDUA.

First, the comparative results obtained by EMPs [42], MEMPs [43], EPF [54], IFRF [61], RGF [67], and TsF [59] demonstrated that image filters can achieve better classification accuracies owing to the image filter that can smoothen the noise of remote sensed images. In addition, the proposed approach can obtain better accuracies than using the above image filters. Because the proposed approach has advantages in preserving the shape and size of different ground targets.

Second, the sensitivity between accuracies and the parameters of T_1 and T_2 is analyzed. Fig. 17 shows that the classification accuracy increases with the increment of T_1 where T_2 is fixed because T_1 denotes the spectral similarity difference between a pixel and its neighboring pixels in the extension progress of an adaptive region. When T_1 is too small, it is insufficient to extend an adaptive region and explore enough contextual information. With the increment of T_1, the size of the adaptive region around each pixel increases, and enough contextual information is utilized. Moreover, owing to the constraint of T_2, the accuracy does not increase continually with the increment of T_1. On the contrary, when T_1 is fixed and T_2 is varied, a similar relationship between T_2 and classification accuracies can be concluded, as shown in Fig. 18.

Third, the relationship between training samples and classification accuracies is investigated. Fig. 19 shows the classifi-

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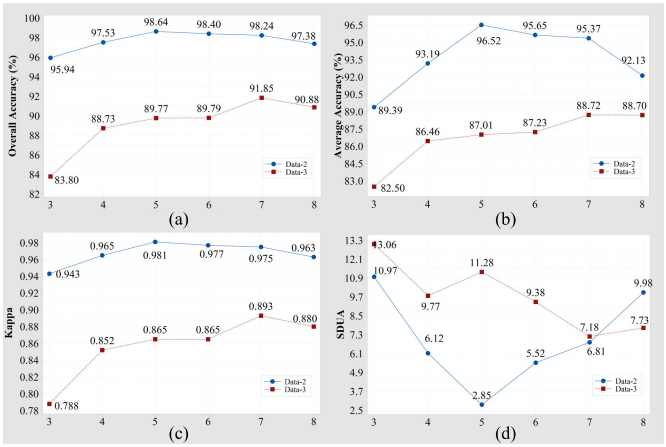


Fig. 17. Relationship between T_1 and classification accuracy when T_2 is fixed at 550 and 400 for Data-2 and Data-3 correspondingly. (a) Overall accuracy, (b) Average accuracy, (c) Kappa coefficient, and (d) SDUA.

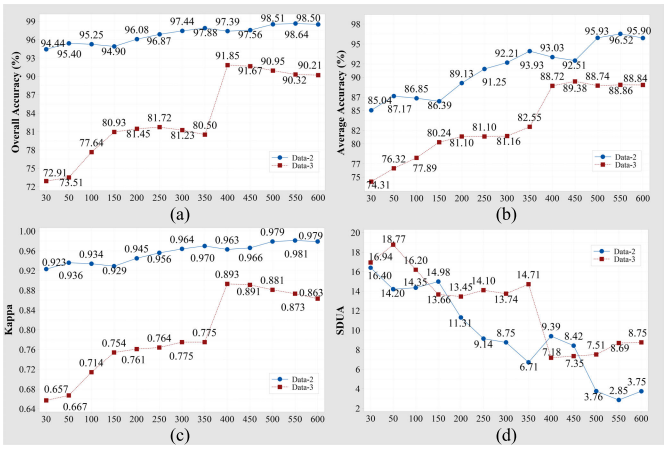


Fig. 18. Relationship between T_2 and classification accuracy when T_1 is fixed at 5 and 7 for Data-2 and Data-3 correspondingly. (a) Overall accuracy, (b) Average accuracy, (c) Kappa coefficient, and (d) SDUA.

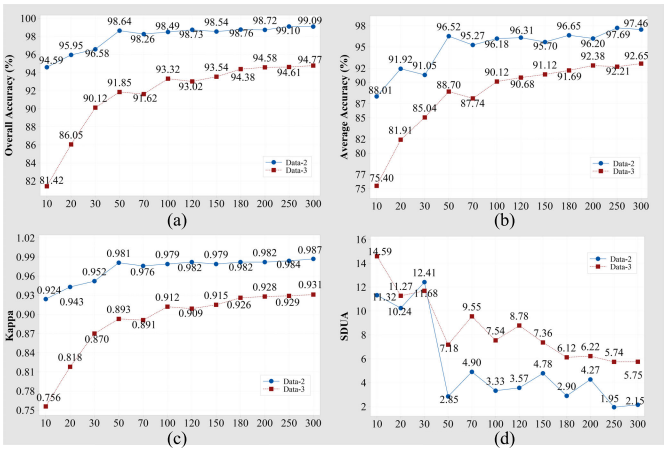


Fig. 19. Relationship between classification accuracies and the number of training samples for the SVM classifier. (a) Overall accuracy, (b) Average accuracy, (c) Kappa coefficient, and (d) SDUA.

cation accuracies first increase with the increment of training samples from 10 to 50 pixels/class, then pose to be a horizontal level, and finally stay at the similar level with the increment of training samples of each class. This finding may be beneficial

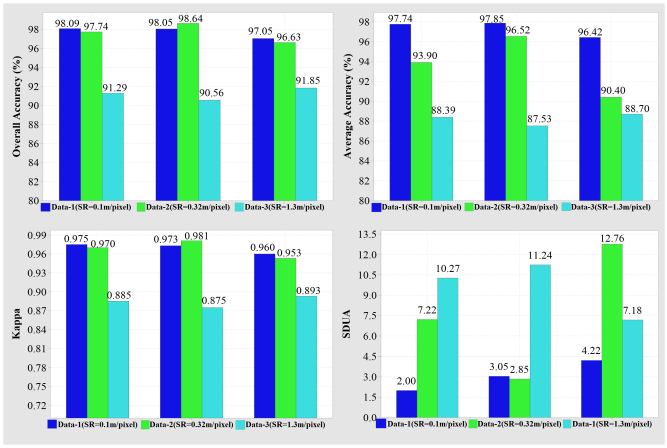


Fig. 20. Investigation between the classification accuracies and the datasets with different spatial resolution. SR is the abbreviation of spatial resolution.

to determine the minor sample requirement of our proposed features in practical applications.

Fourth, the relationship between the resolution and classification accuracies was investigated. As shown in Fig. 20, it indicated that the proposed approach achieved better accuracies with higher spatial resolution remote sensed images if other factors are not considered. Because the image with higher spatial resolution means higher homogeneity of a local area and our proposed approach can preserve the shape and size of the local area and improve the local homogeneity.

The above-discussed conclusions were obtained by applying our proposed features to the Data-2 and Data-3 with SVM classifier. When applying different datasets with different supervised classifiers, the trial-and-error experiments must be conducted to determine the optimal composition of T_1 and T_2, and the minor requirement samples for each class.

IV. Conclusion

In this article, two novel features, i.e., the GWS feature and ASI feature are proposed to improve land cover classification accuracies with HSRRS image. To achieve the objectives, an adaptive region is first suggested to utilize the spatial-contextual information around each pixel. Then, the GWS feature is defined to improve the spectral homogeneity of an adaptive local area around each pixel, and the ASI feature is defined to reveal the relationship between area and shape for an adaptive region around each pixel. Finally, the feasibility and advantages of the proposed novel features are verified by applying them to classification with HSRRS images.

Experiments with four real HSRRS images that are typical urban and rural scenes with various land cover types (trees and meadows, roads, and buildings) are conducted for comparisons. Compared with seven related works and five deep leaning methods, the quantitative evaluation and visual observations indicate that coupling the proposed features with basic RGB features can achieve higher accuracy and balance the user's accuracy for different classes.

In summary, the proposed GWS and ASI features are simple but promising features for HSRRS image classifications. However, they require trial-and-error experiments to obtain the optimal parameters of T_1 and T_2 for a specific image.

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Therefore, our future studies will concentrate on promoting an automatic approach to estimate the optimal composition of T_1 and T_2 for a given image.

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